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Magnetostratigraphy of the Late Triassic Bolücektası Tepe section (southwestern Turkey): implications for changes in magnetic reversal frequency

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ABSTRACT

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A magnetostratigraphic study of light-pink to white pelagic limestones from the Bolücektası Tepe section (southwestern Turkey) provides a magnetic polarity sequence well calibrated with the conodont zonation for the Lower Carnian and uppermost Carnian–Middle Norian (Late Triassic). Thermal (between 400 and 550°C) and alternating field (30–60 mT) demagnetizations isolate the characteristic remanent components. Rock magnetic experiments suggest that magnetite is the dominant magnetic carrier, but a small hematite fraction also exists in the Lower Carnian light-pink limestones. A major sedimentary gap (Middle–Upper Carnian) divides the section in two submembers which reveal two statistically different mean directions, differing in declination by about 30°. A sequence of 39 polarity intervals is obtained, 18 in the Carnian (ten in the Lower Carnian and eight in the uppermost Carnian) and 21 in the Lower–Middle Norian. The magnetic reversal frequencies which can be derived from these results, together with the available magnetostratigraphic data for the Middle–Early Jurassic, Triassic and Late Permian, show a pattern which is in rough agreement with the suggested 150–200 million year time constant in the reversal process since the Kiaman superchron. However, significant short-period fluctuations in reversal frequency cannot be excluded.

1. Introduction

The magnetic reversal time scale is now reasonably well known since the Late Jurassic (160 Ma), mainly from the study of sea-floor magnetic anomalies (for a review see Lowrie, 1989). Analyses of this sequence of about 300 polarity intervals reveal large changes in the frequency of reversals (Lowrie, 1982; Lowrie and Kent, 1983; McFadden and Merrill, 1984). Despite minor differences, these analyses agree in their general conclusions. Reversal frequency has varied in a rather smooth way since the Late Jurassic. The rate of reversals gradually decreased from about 4 reversals per million years to 0 reversals at 124 Ma when a 41-million-year-long normal period occurred, the so-called Cretaceous long normal superchron (LNS). From 83 Ma onward, a monotonic, apparently linear increase brought the rate back to about 5 reversals per million years. A maximum may have been reached 10 million years ago. There have been suggestions that 15- or 30-million-year-long period fluctuations are superimposed on this simple trend (Lowrie, 1982;

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Lowrie and Kent, 1983; Mazaud et al., 1983). However, this observation depends greatly on the method of analysis (sliding time windows). Regardless of the significance of these shorter term fluctuations, the long-term behavior stands out clearly. For preoceanic periods (prior to about 160 Ma, at the end of the Middle Jurassic), the sequence of reversals of the Earth's magnetic field is poorly known. Only a few reliable magnetostratigraphic studies exist; a review is given later in the text. However, a superchron of reversed polarity (LRS), known as the Kiaman superchron, was recognized more than 30 years ago by several authors (Graham, 1955; Cox and Doell, 1960). This superchron occurred in Permo–Carboniferous time, approximately from 320 to 255 Ma.

On the basis of the post-Jurassic trends in magnetic reversal frequency, and by analogy between the Cretaceous long normal interval and the Kiaman reversed superchron, McFadden and Merrill (1984), Courtillot and Besse (1987) and Besse and Courtillot (1991) have suggested a 150–200 million year time constant in the changes of the reversal frequency. Following this model, the Middle–Early Jurassic, for which no continuous magnetic sequence is presently available, would correspond to a period of high frequency and the Triassic to a period of medium frequency with a marked increase between the Early and Late Triassic. Many models have been suggested to explain the origin of changes in reversal frequency (Jones, 1977; McFadden and Merrill, 1984, 1986; Loper and McCartney, 1986; Courtillot and Besse, 1987; Larson, 1991; Besse and Courtillot, 1991). These variations are generally explained in terms of thermal changes at the core–mantle boundary, possibly within the D″ layer, and have been associated with long-term variations of core, mantle, and lithosphere dynamics. Common time constants on the order of 150–200 million years have been suggested for these variations (for a review, see Courtillot and Besse (1987)). However, most of these models remain untested. In this context, magnetostratigraphic studies are essential to pursue the analysis of the reversal behavior over longer time scales. In this paper, we investigate the magnetostratigraphy of a Late Triassic pelagic limestone section from southwestern Turkey. We present a new magnetic sequence well calibrated with the conodont zonation for the Lower Carnian and the uppermost Carnian–Middle Norian. An estimate of reversal frequency for the Late Triassic derived from our new data, along with the published magnetostratigraphic data of Middle–Early Jurassic and Early Triassic–Late Permian age, give a first indication of the magnetic reversal frequency behavior since the Kiaman superchron.

2. Geological setting, lithostratigraphy and sedimentology

The studied section is located in the southwestern part of Turkey (Fig. 1). The rocks described in the following paragraph are widely distributed through the Taurides and form a distinctive part of the Triassic sequence within the so-called Antalya calcareous nappes (ACN) or Upper Antalya Nappe (Lefèvre, 1967). These nappes belong to the western Taurus structural system (Blumenthal, 1960; Brunn et al., 1971, 1976; Gutnic et al., 1979), cropping out now to the south of the Taurus calcareous axis (Ricou et al., 1975). These sediments, which may represent deposition on a distal pelagic plateau, are also known from many areas within the Tethyan region, for example from the calcareous Alps of Austria where they originally were named, about 150 years ago, as Hallstatt limestones.

The investigated section belongs to the Bakırlı Dag units of the ACN (Delaune-Mayere et al., 1977; Marcoux, 1979; Ricou et al., 1985) and is located on the east side of the Tepe dag, 'dag' meaning mountain in Turkish, near a small, well-defined summit called Bolücektası Tepe. The sequence is basically monoclinal, dipping gently to the west. The dip slightly increases from bottom to top (dipping from 25 to 40° toward the west) and there is a change in the strike direction (180–197°) between the Lower Carnian and the Upper Carnian parts of the section as a result of recent faulting. The main trends of the lithostratigraphy are described in Marcoux (1979, 1987), Argyriadis et al. (1980) and Marcoux et al. (1986,

1987). Four successive members are distinguished within the Bolücektası Tepe section (Fig. 2). These members reflect a continuous development from a highly oxidized and condensed pelagic plateau site during the Upper Ladinian and Lower Carnian (member 1) to a more continuous sedimentation regime during the uppermost Carnian and Lower Norian (members 2 and 3). The uppermost member (4) of Middle Norian age was influenced by prograding of shallow-water debris. The increased influx of sediment progressively filled the basin, with shallow-water limestones of Dachstein type ('Bakırlı Dag' Formation) occupying the topmost Triassic.

2.1. Nodular member (1)

This member has a thickness of about 9 m and is predominantly constituted of wavy to nodular, thin-bedded, brownish-red limestones with occasional intercalations of continuous, even-bedded limestones of 10–20 cm thickness. The latter contain ferromanganese crusts and nodules resulting in dark-colored bedding surfaces. These beds are important because they contain corroded fragments of ammonites, the only extractable ones found in the whole section. Some of these beds are also topped by a thin iron–manganese cover with encrusted organisms such as crinoid roots, serpulids and sessile foraminifera ('Tolypaminna') which indicate breaks in sedimentation.

This facies, which is similar to the Tethyan Ammonitico Rosso, is characterized by (1) a highly oxidizing environment, (2) a slow sedimentation rate, (3) early lithification, (4) winnowing and reworking of sediment by strong episodic currents, and (5) formation of stratigraphically condensed ammonite accumulations during phases of stagnant deposition.

2.2. Bedded member (2)

After a major break in sedimentation (Fig. 2), the nodular member is discontinuously overlain by about 10 m of even-bedded, light-colored, fine-grained limestones. The thickness of individual beds increases from about 30 cm to 1 m at the top of this member. A single red-colored bed with iron–manganese crust, similar to that observed in the nodular member, occurs close to the base of this member. Three stratigraphically important halobia-bearing levels are also noted (1.0, 3.5 and 8.0 m from the base, Fig. 2). By comparison with the underlying member, the sedimentation rate is faster and more continuous with a progressive transition toward the overlying massive member.

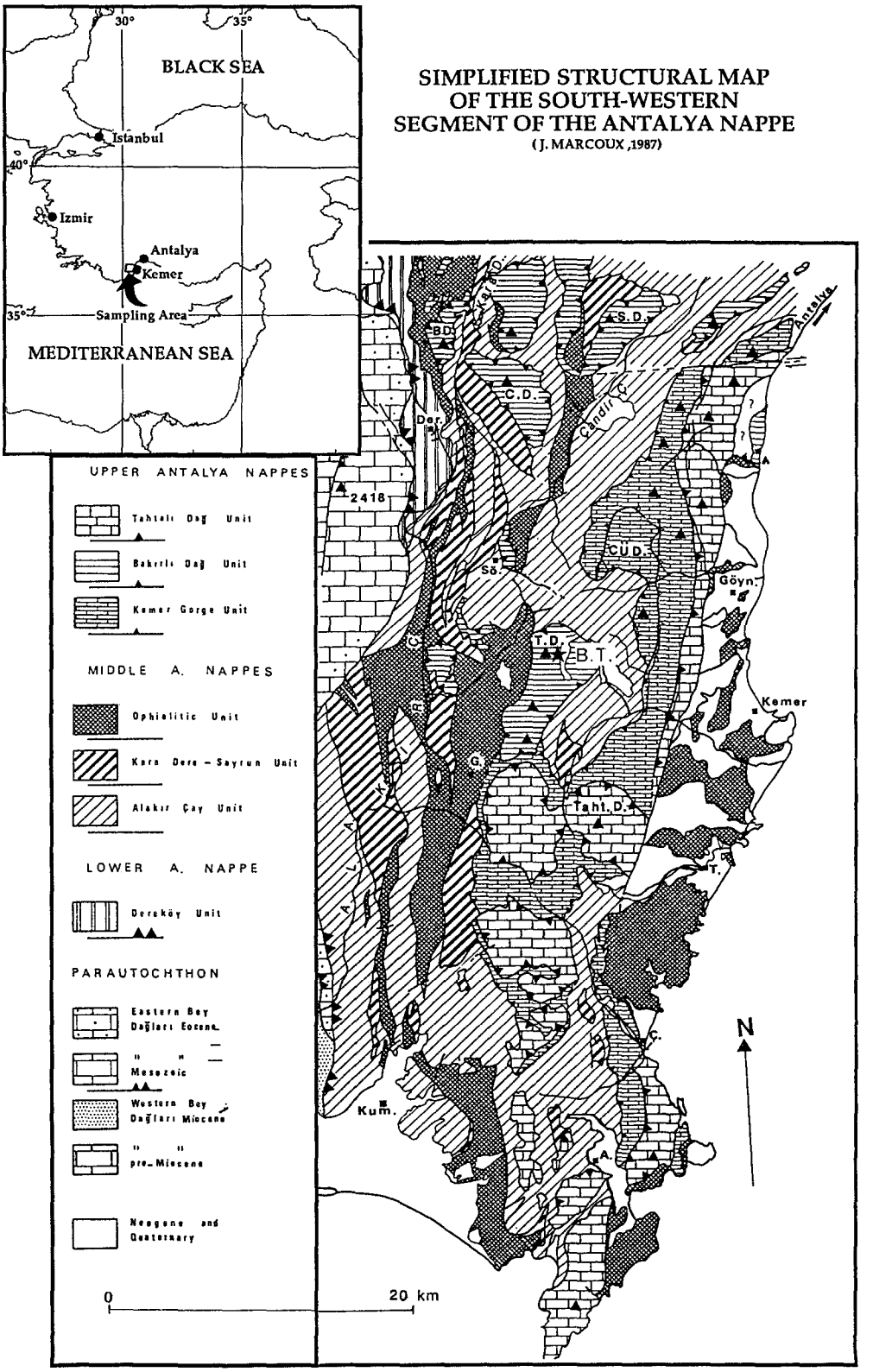
2.3. Massive member (3)

The light-colored cliff-forming limestone is not strictly massive but contains indistinct bedding on a 2–3 m scale. Well-defined bedding planes are rare and are limited to the top of the sequence. This 24 m thick member is significantly affected by faulting which, together with the uniformity of the sequence, hampers an exact restoration of this part of the section. For example, the main fault at the top of this member (Fig. 2) was primarily recognized by biostratigraphy and only subsequently identified in the field. Its uniform, more or less unstratified character points to an increasing and continuous sedimentation rate.

2.4. Bedded allodapic member (4)

This member is located at the top of our sequence with a thickness of more than 35 m. It consists of whitish to subordinate pinkish limestone beds of variable thickness (1–2 m) except for two unbedded intervals in the middle and upper parts. The most characteristic, and therefore, name-giving, feature is the presence of coarser grained allochthonous layers composed of carbonate debris within the dominating fine-grained autochthonous matrix. The upper half of the sequence is further distinguished by the presence of stromatactis-like cavities reaching 10 cm in length and 1–3 cm in height.

This member most likely reflects an identical basinal setting as in member 3, but one which became strongly affected by debris input from a nearby carbonate platform and/or from a shallow-water area. This sediment supply may have resulted in an increase of the sedimentation rate


SIMPLIFIED STRUCTURAL MAP
OF THE SOUTH-WESTERN
SEGMENT OF THE ANTALYA NAPPE
(J. MARCOUX, 1987)
30°
35°
BLACK SEA
Istanbul
40°
Izmir
Antalya
Kemer
35°
Sampling Area
MEDITERRANEAN SEA
UPPER ANTALYA NAPPES
Tahtalı Dağ Unit
Bakırlı Dağ Unit
Kemer Gorge Unit
MIDDLE A. NAPPES
Ophiolitic Unit
Kara Dere – Sayrun Unit
Alakır Çay Unit
LOWER A. NAPPE
Derekôy Unit
PARAUTOCHTHON
Eastern Bey Dağları Eocene
" " Mesozoic
Western Bey Dağları Miocene
" " pre-Miocene
Neogene and Quaternary
0
20 km
2418
B.D.
S.D.
C.D.
Der.
Çandır
Antalya
Sö.
C.U.D.
Gövn.
T.D.
B.T.
Kemer
Tahl.D.
T.
Kum.
A.
N

and in a more complete sequence during the time interval represented in this member.

3. Biochronology

High-resolution biochronologic dating of Triassic rocks is only possible with ammonoids, conodonts or, to a lesser extent, with pelagic bivalves. For the Carnian, the integration of conodont zones with ammonoid zones is based on Krystyn (1980, 1983). For the Norian, only a tentative correlation was achieved using Orchard's (1983) data and unpublished data of one of us (L. Krystyn) from ammonoid-rich 'Hallstatt' limestones from Timor.

In the Bolücektası Tepe section, ammonites and halobiids are restricted to single widely spaced beds and all biostratigraphic boundaries are, therefore, based on conodont data. For this purpose, more than 60 samples (with a total weight of 130 kg) have been collected giving the following zonation (Fig. 2). The upper part of the reddish nodular member (1) represents the Julian 1 and 2 conodont zones of the Lower Carnian. Just above, a large sedimentary gap exists, since the Tuvalian 1, 2 and Tuvalian 3-I ammonoid zones are missing. Renewal of sedimentation occurred during the Tuvalian 3-II in the uppermost Carnian (bedded member 2). The Lower Norian is represented by the Lacian 1 (upper part of member 2) and 2 (massive member 3) ammonoid zones but the Lacian 3 zone is missing. The Middle Norian (bedded allodapic member 4) is represented by the Alaunian 1, 2 and 3 (only the lower part) ammonoid zones.

4. Triassic chronology

A prerequisite for the establishment of a reliable magnetostatigraphic time scale and for the precise estimation of the reversal rate is the availability of accurate radiometric age data from rocks whose stratigraphic positions are well defined. Although it is beyond the scope of this paper to make an extensive review of the radiometric data, some consideration of the age of the chronostratigraphic boundaries classically available in the literature is given. In general (i.e. not only for the Triassic), data controlling the ages of chronostratigraphic boundaries are sparse and unevenly distributed. Most of the radiometric ages come from remote stratigraphic units which cannot be unambiguously correlated, emphasizing the importance of a global biostratigraphic zonation. In addition, radiometric age data rarely correspond to biostratigraphic boundaries and require interpolation.

On Fig. 3 we have plotted the numeric ages of the Triassic chronostratigraphic boundaries with their uncertainties according to three published time scales: Foster and Warrington (1985), Harland et al. (1989) and Odin and Odin (1990). Estimation of errors is different for each author. For example, Forster and Warrington (1985) give a somewhat arbitrary error limit of ± 5 million years. In contrast, Odin (1982) and Odin and Odin (1990) simply report the range of reliable ages near a given boundary (when the data are available), providing an interval within which the boundary lies. Harland et al. (1982, 1989) use the method of chronograms: the age of a limit is given by the minimum of an error function in which selected data are ponderated according to their 1σ experimental error bars. A quality factor is given for each chronogram, and only the best ones are used as tie-points. The ages of the boundaries devoid of reliable radiometric data are given by interpolations between tie-points.

For the Upper Triassic, the Harland et al. (1989) time scale clearly differs from the other two time scales. Most of the differences result from the data selection. The Jurassic–Triassic boundary is reasonably well defined and can be confidently assigned to an age of 205 Ma ± 4 following Odin and Odin (1990) or Foster and

Fig. 1. Geologic map (from J. Marcoux, 1987) of the Antalya region (southwestern Turkey). Black triangles indicate mountain summits. The full star indicates the location of the Bolücektasi Tepe section (B.T.).

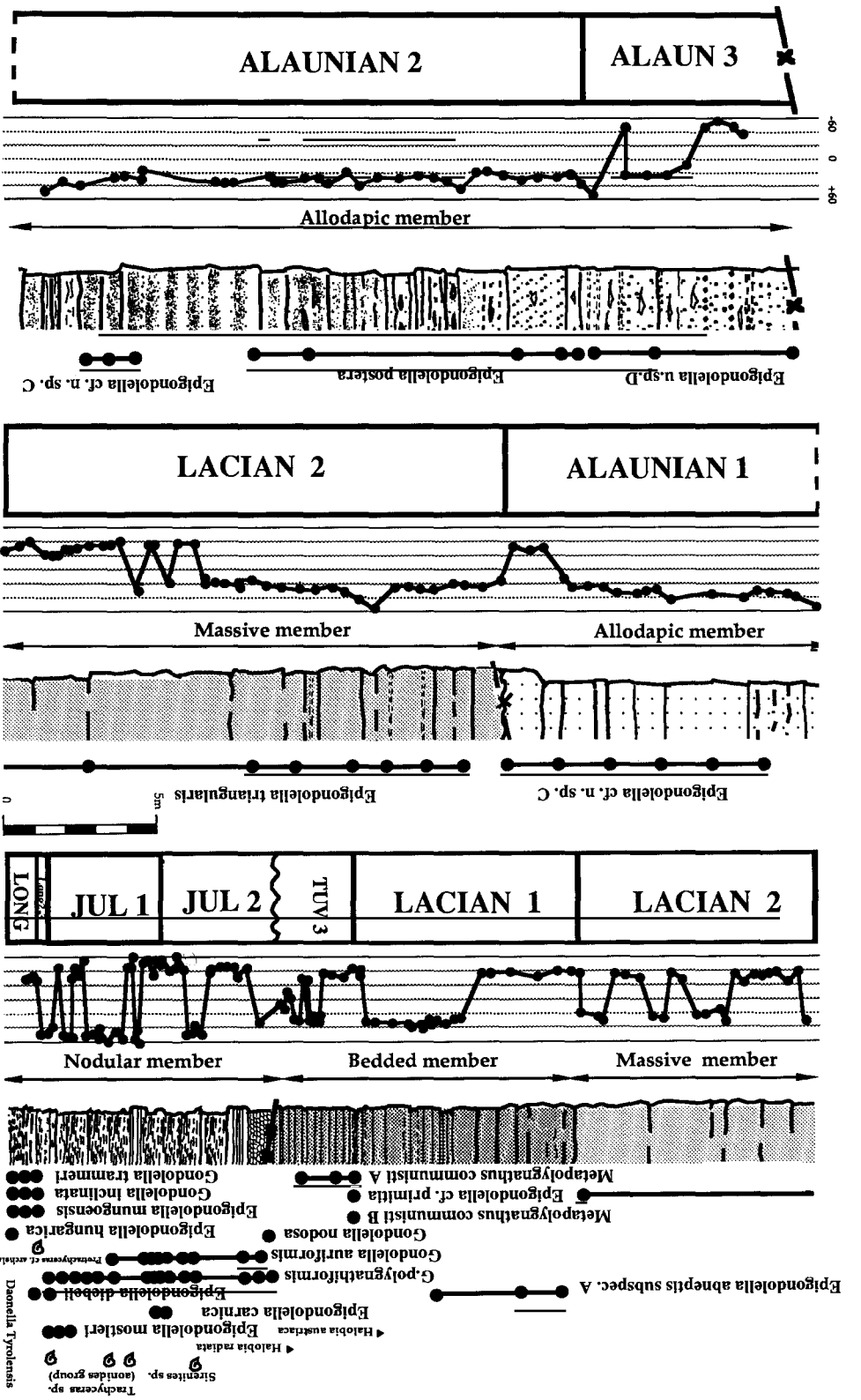
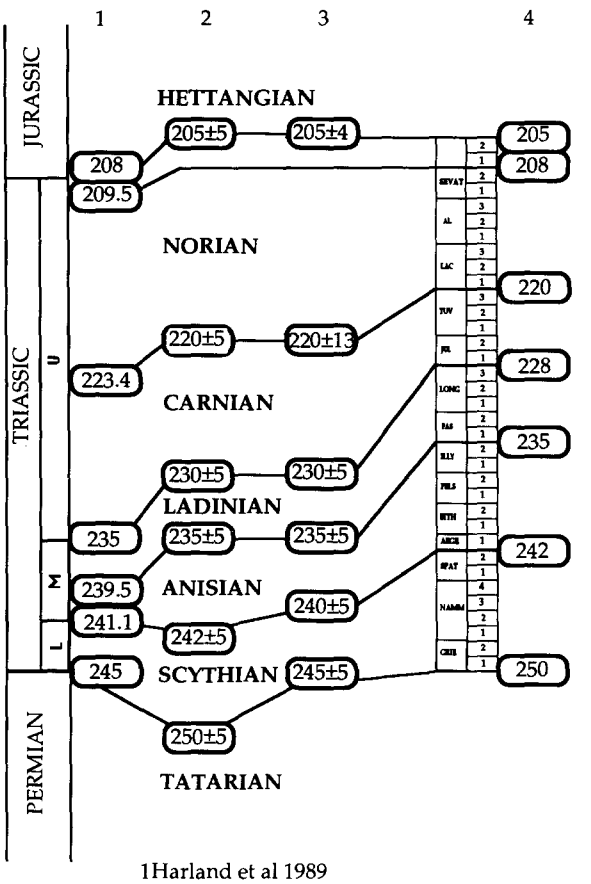


Fig. 2. Lithostratigraphic log and biozonation of the Boliücektasi Tepe section. Four successive members can be distinguished reflecting a continuous development from a highly oxidized and condensed pelagic plateau site during the Upper Ladinian and Lower Carnian (first member) to a more continuous sedimentation system during the uppermost Carnian–Middle Norian. A detailed description of the biostratigraphy is given in the text.

Warrington (1985) and to a nearly identical value of 208 Ma according to Harland et al. (1989). The duratation of the Rhaetian (topmost Triassic) is still a matter of debate, but could have been between 1 and 4 million years. The Carnian–Norian boundary is poorly defined, in the range 212–238 Ma (with a rough 'best' estimate at 220 Ma ± 13) following Odin and Odin (1990), 220 Ma ± 5 according to Foster and Warrington (1985), and a nearly identical value of 223 Ma by interpolation from Harland et al. (1989). Among the Triassic stage boundaries, the best one is certainly the Aniso–Ladinian boundary, which is confidently located between 229 and 237 Ma ('best' estimate at 235 Ma ± 5) following Odin and Odin (1990), 235 Ma ± 5 for Forster and Warrington (1985), and 239 Ma (minimum 233, maximum 246) for Harland et al. (1989). Despite a discrepancy of about a 5 million year shift between the three time scales, the total duration of the Upper Triassic is nearly identical for all the authors. In view of these uncertainties, the duration of the Upper Triassic would be 25 million years (maximum 35, minimum 16) following Forster and Warrington (1985) and Odin and Odin (1990), and 27 million years (maximum 37, minimum 16) for Harland et al. (1989).

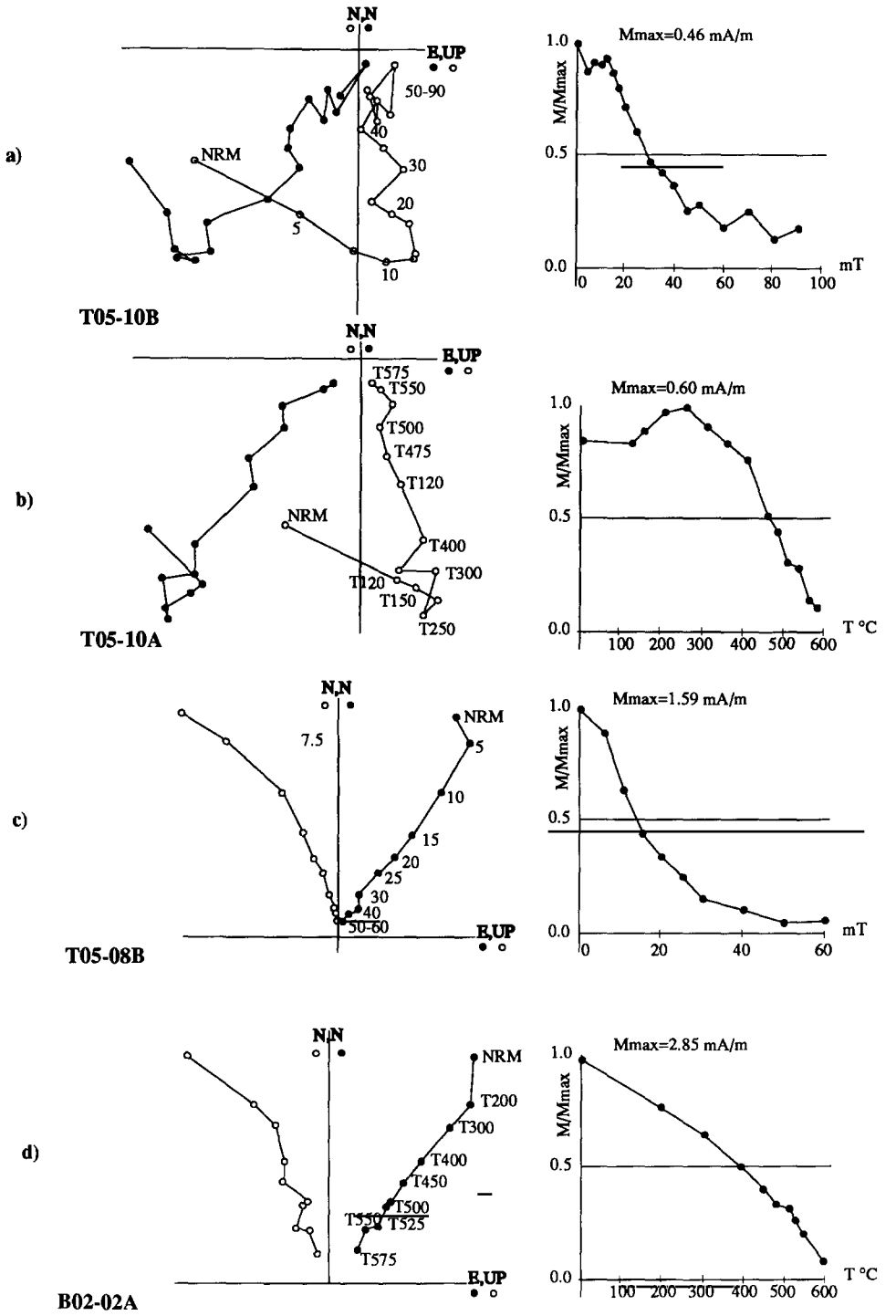
The large uncertainties observed above are due to the paucity of radiometric data for the Triassic, and the age of the limits can only be considered as tentative. An alternative, but somewhat extreme procedure for estimating the age of the Triassic stage boundaries is to assume an equal duration for ammonoid zones, a method commonly used (but not satisfactorily justified) by most stratigraphers and paleontologists (Van Hinte, 1978; Westerman, 1984). Following Forster and Warrington (1985), and considering the recent magmatic age estimate of 251 Ma ± 3.4 (2σ) from Claoué-Long et al. (1991), we decided to date the base of the Triassic at 250 Ma. We also consider the Jurassic–Triassic boundary at 205 Ma and the Aniso–Ladinian boundary at 235 Ma (Forster and Warrington,1985; Odin and Odin, 1990). Over the two segments between (1) the Rhaetian and the Ladinian and (2) the Anisian and the Scythian, 35 standard ammonoid zones are distinguished within the Tethyan Triassic (Fig. 3; L. Krystyn, personal communication), giving a mean value of 1.5 million years per ammonoid biochron for the first segment and 1.0 million



1 Harland et al 1989
2 Forster and Warrington 1985
3 Odin and Odin 1990
4 Equal chronozones

Comparison of the numeric ages for the Triassic

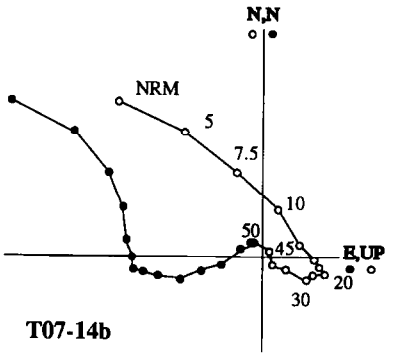
Fig. 3. Comparison of three Triassic time scales (Forster and Warrington, 1985; Harland et al., 1989; Odin and Odin, 1990). The duration of the Upper Triassic is quite similar for these times scales (about 25 Ma), but a shift in age is clear between the Harland et al. (1989) time scale and the two others. An estimation of the numerical age, using the concept of equal duration for the 35 ammonoid zones interpolated between three tie-points (Jurassic–Triassic, Aniso–Ladinian and Permo–Triassic) distinguished through the Triassic, is also shown on the right side of the figure.

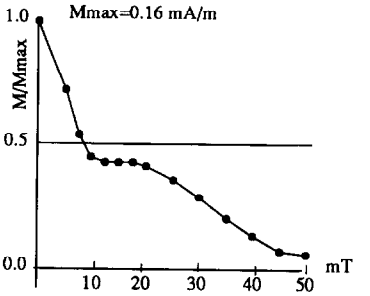


All diagrams are in stratigraphic coordinates.

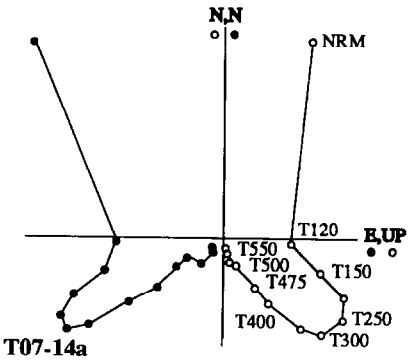
Fig. 4. Examples of orthogonal projections of remanent magnetization vector changes during thermal ((b), (d), (f), (h), (i) and (j)) and AF demagnetization ((a), (c), (e) and (g)) for some representative samples after bedding correction (left side). The open (close) symbols refer to the horizontal (N–S vertical) plane. We also indicate the respective normalized change in the intensity during the demagnetization (right side). (a)–(d) Reddish limestones (Lower Carnian); (e)–(j) whitish limestones (uppermost Carnian–Middle Norian).

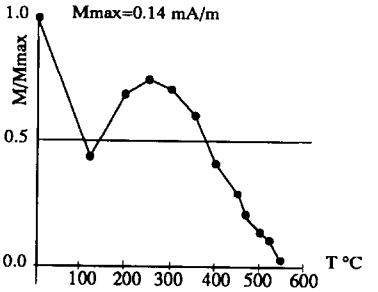
e)



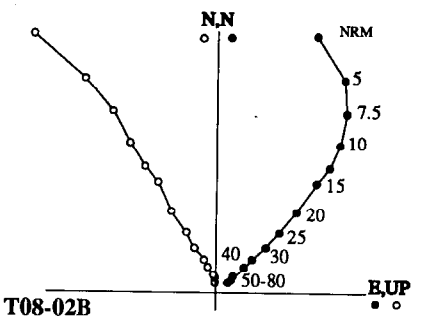


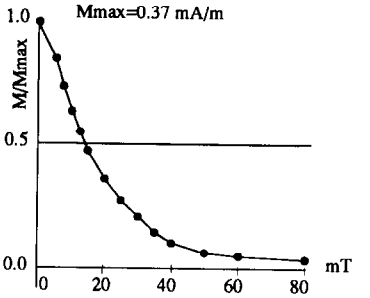
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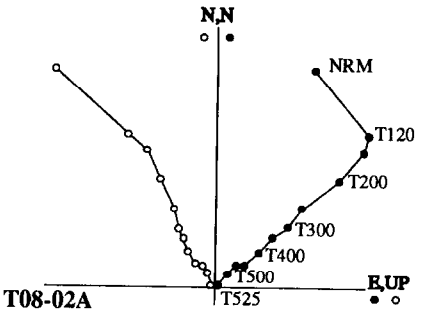


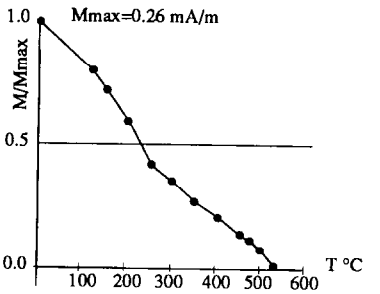
g)





h)





All diagrams are in stratigraphic coordinates.

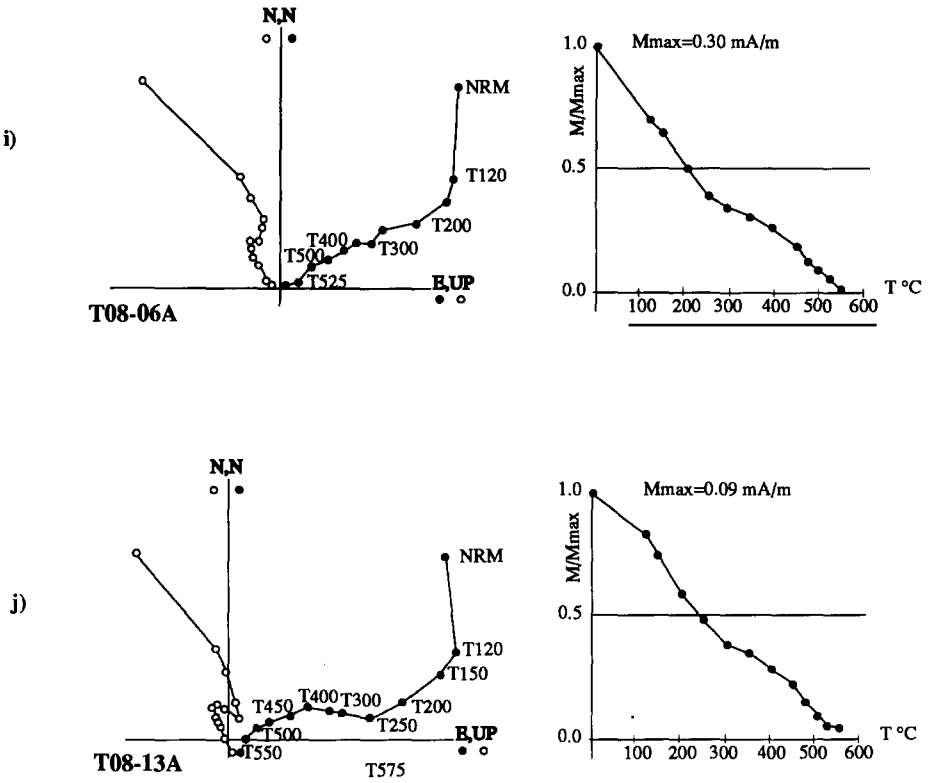


Fig. 4 (continued).

years per ammonoid zone for the second. When the new age estimates for the stage boundaries (Fig. 3) are compared with the other published time scales, some major differences are noticeable. The relatively short duration of the Late Triassic in this estimate (23 million years; 27 million years in Harland et al. (1989)) is balanced by a duration of 14 million years for the Middle Triassic (6 million years in Harland et al. (1989) and 10 million years in Odin and Odin (1990)) and 8 million years for the Lower Triassic (4 million years in Harland et al. (1989) and 5 million years in Odin and Odin (1990)). A longer duration of the Early Triassic in respect to the Late Triassic has also been suggested by Odin and Odin (1990). Within the Late Triassic, the age of the Carnian–Norian and Norian–Rhaetian boundary can be fixed at 220 Ma and 208 Ma (respectively, 223.4 Ma and 209.5 Ma in Harland et al. (1989)) giving a duration of 12 million years for the Norian and 8 million years for the Carnian (respectively, 13.9 million years and 11.6 million years in Harland et al. (1989)). These differences in age and duration of the Triassic stages between the time scales will greatly influence the estimate of the magnetic reversal frequency from the magnetostratigraphic data and a comparison using the different time frames will be given.

5. Paleomagnetic analysis

Paleomagnetic analyses have been conducted on 262 samples using a CTF 3-axis SQUID magnetometer in the mu-metal/steel magnetically shielded laboratory at the Institut de Physique du Globe de Paris (IPGP).

The NRM intensity differs significantly between the Lower Carnian light-pink limestones,

with a mean of 0.88 mA $m^{-1} \pm 10\%$, and the whitish limestones of uppermost Carnian–Middle Norian age with a mean of 0.34 mA $m^{-1} \pm 9\%$. The highest intensities correspond to the lowest sedimentation rates.

A pilot study undertaken on 25 samples, with a large number of thermal and alternating field (AF) demagnetization steps (approximatively 15 steps), allowed us to develop a routine paleomagnetic analysis scheme for the remaining samples (about ten steps for the thermal demagnetization). Measurements of magnetic susceptibility have also been performed during the thermal treatment. A lack of noticeable change in susceptibility, which remains very weak with positive and negative values, seems to indicate a high stability of the magnetic mineralogy.

Several cores yielded two specimens, allowing direct comparison of AF and thermal demagnetization (Fig. 4). The different magnetic components can usually be easily isolated both by thermal and AF demagnetization even though the demagnetization paths in the lower part of the section are more scattered. The general demagnetization characteristics show that an initial soft component of normal polarity is removed by the first steps of thermal (about 200°C) and AF (about 10–20 mT) demagnetization. The characteristic magnetic polarity is isolated from about 300°C or 20 mT and the characteristic magnetic component is consistent in the 400–575°C temperature range or 30–60 mT AF range. The Lower Carnian samples (light-pink colored limestones) exhibit only two magnetic components, a component roughly parallel to the present field and the characteristic direction (Fig. 4(a)). A clear paleomagnetic behavior is also observed in the samples of whitish limestones (Figs. 4(b) and(c)). However, a more complex signature is sometimes exhibited for samples from this lithology. Indeed, some samples with a normal-polarity characteristic component show a small intermediate component of reversed polarity in stratigraphic coordinates (Fig. 4(c)). This component, which is never properly isolated, affects the demagnetization path to different degrees (compare the samples T08–02a, Fig. 4(b), and T08–06a T08–13a, Fig. 4(c)). Despite this combination of components, samples which have a problematic demagnetization behavior, i.e. no final direction can be confidently obtained, are rare (22 samples). In these cases, both analytical procedures are ineffective, but even if the precise magnetic direction is uncertain, there is no doubt about the assignment of magnetic polarity.

Together with the observed unblocking temperature and AF spectra of the magnetic compo-

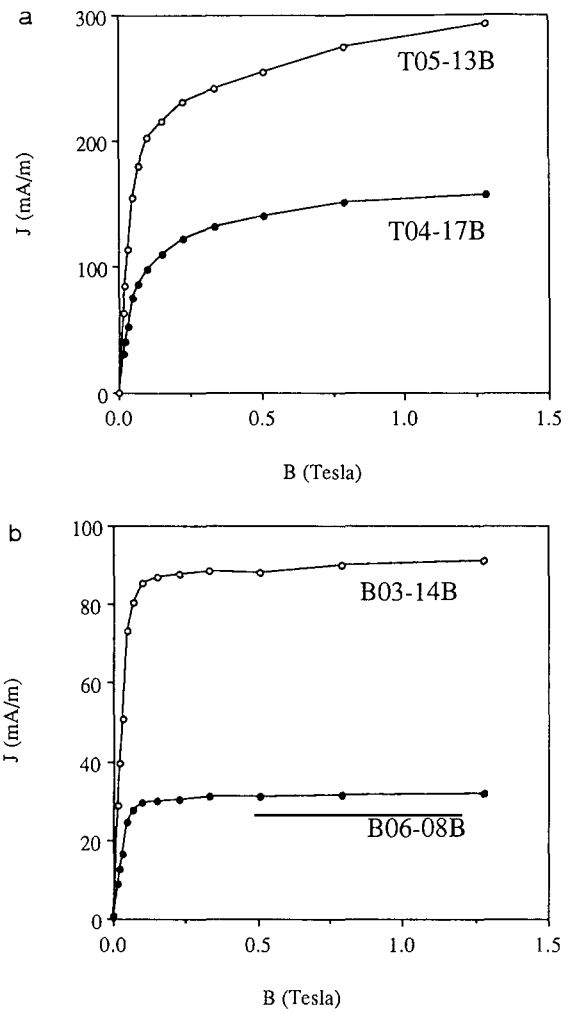


Fig. 5. Acquisition of a monodirectional IRM in samples of both lithologies (a) and (b). The dominant magnetic carrier is magnetite, but an increase of the IRM in the range 0.4–1.4 T clearly indicates a weak fraction of high-coercivity mineral (probably hematite) in the light-pink limestones of Lower Carnian age (a).

nents, the results of an isothermal remanent magnetization (IRM) analysis on selected samples distributed throughout the section, which shows that saturation achieved in fields of about 0.2 T, are in agreement with a magnetic mineralogy dominated by magnetite (Fig. 5). However, a slight increase is observed at fields higher than 0.2 T for the light-pink limestone samples (Fig. 5(a)). To characterize better the magnetic carriers, we have thermally demagnetized a composite IRM as described by Lowrie (1990). On the same core, an IRM of different coercive force (0.12, 0.4 and 2.3 T) is applied successively to one of the three axis and then thermally demagnetized. The results are shown in Fig. 6. The light-pink limestones (Figs. 6(a) and (b)) reveal a magnetization dominated by magnetite with a medium–low coercivity (the low coercivity is predominant). As observed in the classic IRM (Fig. 5), minerals with high coercivity, probably hematite, are present. Thermal demagnetization indicates that these minerals have an unblocking temperature close to 580–600°C. As suggested by Channell (1978) and Channell et al. (1982), dehydration of goethite appears most likely as the origin of formation of this hematite. The whitish limestones (Figs. 6(c) and (d)) show a magnetic mineralogy mostly carried by a low-coercivity mineral, in agreement with the behavior displayed on Fig. 5, and an unblocking temperature of about 550°C. The magnetic carrier is thus probably a titanomagnetite. A small magnetic fraction with high coercivity can be observed in Fig. 6(d). It is removed after the first thermal steps (100–150°C)

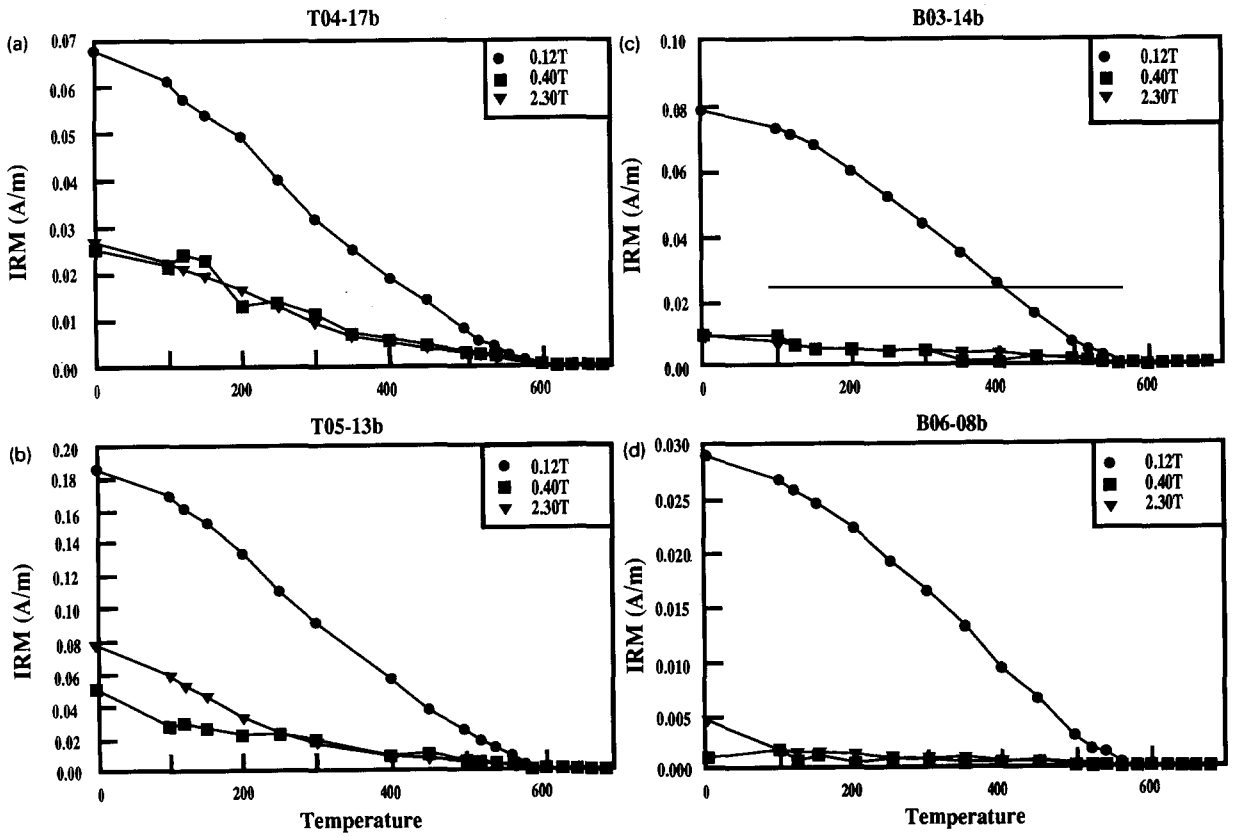


Fig. 6. Thermal demagnetization of a triaxial differential IRM (0.12, 0.4 and 2.3 T; method described in Lowrie, 1990). These curves indicate a magnetization mainly carried by magnetite of medium–low coercivity. A mineral of high coercivity, probably hematite, with an unblocking temperature of about 560°C is recognized in the reddish limestones (a) and (b).

and is thus probably goethite. This contribution has also been identified by joint thermal and AF demagnetization of specimens from the same core (samples T07–14a,b Fig. 4(b)): a component which is removed after the first thermal steps (120 or 200°C) remains after AF demagnetization.

6. Paleomagnetic results

6.1. Mean directions

The characteristic remanent magnetization directions (ChRM) isolated for all samples are well grouped and only rare directions deviate greatly from the mean (Figs. 7(a)–(c); Table 1). The normal and reversed directions obtained for the Lower Carnian have a Fisherian distribution (Fisher, 1953; Fig. not shown) and are antipodal (Fig. 7(c)). The mean direction including all these directions (the distribution is also Fisherian) is defined with a good precision ($a_{95} = 3.6°$). The case is not so favorable for the directions of uppermost Carnian–Middle Norian (Fig. 7(b)). These data depart from a circular symmetry with some dispersion in declination for the normal directions. We suggest that this distribution is biassed by the intermediate reversed overprint shown during the demagnetizations. Demagnetization diagrams (Fig. 4(c)) indicate that this small component can induce an error in the estimate of the declination. Similarly, the superposition of components could explain the slightly non-antipodal character of the normal and reversed (Fisherian) mean directions (Fig. 7(c)).

The two general mean directions (Lower Carnian and uppermost Carnian–Norian) are statistically different ($dD = 28.3° \pm 4.0°$, $d\lambda = 4.8° \pm 2.1°$). We have calculated the mean declinations and mean paleolatitudes of subsites vs. depth using successive windows defined by eight sampled levels (Fig. 8). No clear trend can be observed within the two sequences and all the movement occurs during the sedimentary gap in the Middle–Late Carnian. Geologic observations in the field suggest a fault parallel to the lithification between the two members. Further paleomagnetic studies on others sites are presently being carried out to test a possible synsedimentary tectonic rotation of this part of the Taurus.

6.2. Magnetostratigraphic results

Figure 9 summarizes the variations of NRM intensity and of the ChRM directions. Large fluctuations are observed between 0 and 18 m (mostly Lower Carnian), and seem to be gradually reduced in the uppermost Carnian, becoming very weak in the Lower and Middle Norian. A clear correlation exists between the NRM intensities for the light-pink limestones and the final polarity after demagnetization, the weaker NRM corre-

TABLE 1

Mean paleomagnetic directions and VGP of the Bolücektasi Tepe section (stratigraphic coordinates)

	No. Samples	Declination	Inclination	alp95	k
Lower Carnian					
Normal	26	31.5°	14.5°	5.1°	31.5
Reverse	34	211.6°	−8.3°	5.0°	25.5
N+R	60	31.0°	11.0°	3.6°	27.1
VGP	Latitude: 48.0°N; longitude: 160.6°E; dp: 1.9°; dm: 3.7° Paleolatitude: 5.6°N				
Uppermost Carnian – Middle Norian					
Normal	116	60.0°	18.1°	1.9°	46.8
Reverse	56	237.6°	−24.1°	3.1°	39.4
N+R	172	59.3°	20.1°	1.7°	41.8
VGP	Latitude: 30.7°N; longitude: 130.9°E; dp: 0.9°; dm: 1.8° Paleolatitude: 10.4°N				
Difference 1/2					
dD: 28.3°±4.0°			dpaleolat: 4.8°±2.1°		

N, normal; R, reverse.

sponding to reversed polarity. This can be explained easily by the superimposition of a normal low-blocking temperature or low-coercivity component, destroyed during the demagnetization process. The mean intensity of the samples of normal polarity (1.18 mA $m^{-1} \pm 12.0\%$) is twice the mean intensity of the samples of reversed polarity (0.62 mA $m^{-1} \pm 8.0\%$). The difference is

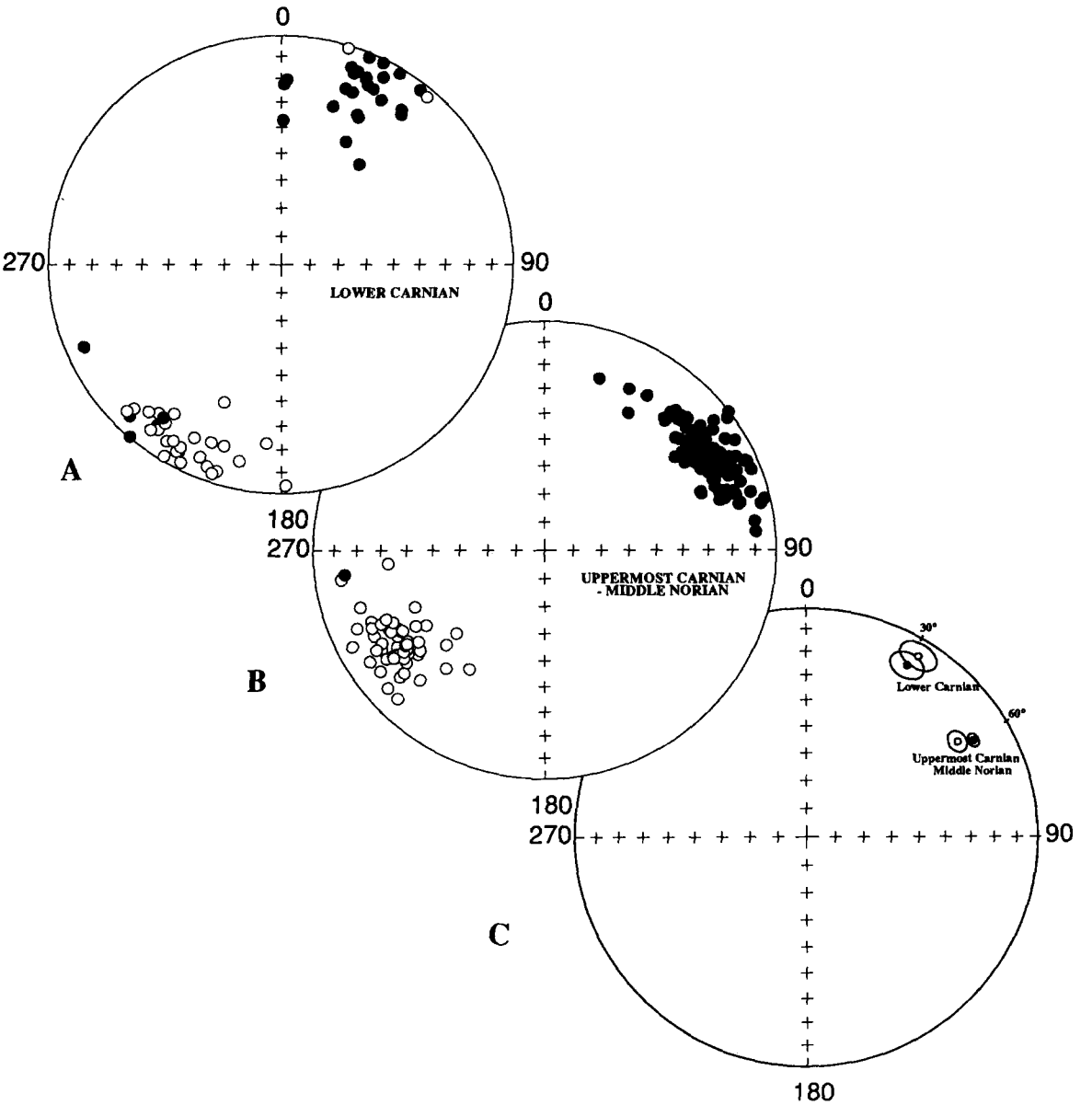


Fig. 7. Equal-area projection of the ChRM directions after tectonic correction (a) and (b) and their mean directions (c). Full (open) dots represent the projection of the directions in the lower (upper) hemisphere. (a), (b) Directions observed in the Lower Carnian sequence and uppermost Carnian–Middle Norian sequence, respectively. (c) Mean directions obtained in the Bolücektasi Tepe section.The two reversed mean directions (open circles) have been inverted for comparison with the respective normal directions (full circles).

not as great in the whitish limestones, respectively 0.38 mA $m^{-1} \pm 8.3\%$ and 0.26 mA $m^{-1} \pm 9.4\%$.

The ChRM clearly define a magnetic zonation. The reversals are most evident in declinations while the low inclinations show a less interpretable signal. We have computed the virtual geomagnetic pole (VGP) latitude for each direction and the values are interpreted in terms of normal and reversed polarity (Fig. 10). The Carnian exhibits 18 polarity zones, ten in the Lower Carnian (Julian 1 and 2 conodont zones) and eight in the Tuvalian 3II zone (uppermost Carnian); four magnetic intervals defined by only one sample correspond to condensed sedimentary layers (small horizontal lines in Fig. 10). One of these contains the Middle–Late Triassic (Ladinian–Carnian) boundary. A fifth poorly defined zone corresponds to the break in sedimentation during the Middle–Upper Carnian. De-

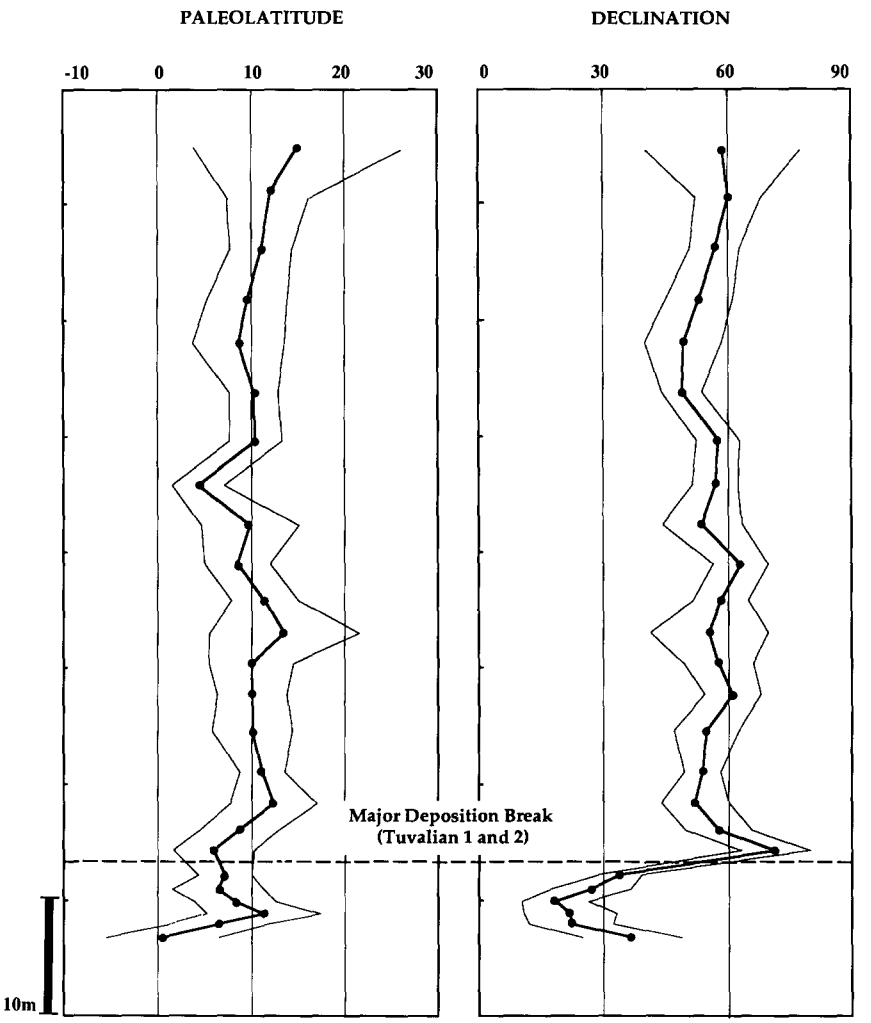


Fig. 8. Stratigraphic plot of mean paleolatitudes and mean declinations estimated from subsites of eight independent levels. Dashed lines indicate a major break in deposition. The error in paleolatitude is given by dp (with $dp = 0.5a_{95}(1+3\cos^2 p)$ where p is the ancient colatitude, e.g. McElhinny, 1973). The error in declination is $a_{95}/\cos I$ (e.g. McElhinny, 1973). No apparent trend is observed within the two sequences.

spite these intervals, a clear magnetostratigraphic sequence is obtained. For the Lower and Middle Norian (Lacian 1–Alaunian 3 conodont zones), 21 magnetic intervals are obtained but four of them are again defined by a single sample.

To estimate the magnetic reversal frequency from these results, we must now appraise the completeness of the Bolücektası Tepe section. For the Carnian, the conodont Julian 1 and 2 zones are complete albeit that the existence of minor gaps is probable. The Tuvalian 1 and 2 and part of Tuvalian 3 are missing. Thus, not much more than half of the Carnian is exposed, representing 6 million years in duration following Harland et al. (1989). For the Norian–Rhaetian, five conodont zones of ten are present although one (the Alaunian 3 zone) is incomplete. Taking an approximate duration of 15 million years for this period (Harland et al., 1989), about 7 million years are represented in our section. From these rough estimates, a 'first-order' magnetic reversal frequency can thus be derived, giving a value of about 2.8 reversals per million years for the Carnian and 2.9 reversals per million years for the Norian. Excluding the intervals represented by a single sample, these estimates become 1.8 reversals per million years and 1.9 reversals per million years for the Carnian and Norian, respectively.

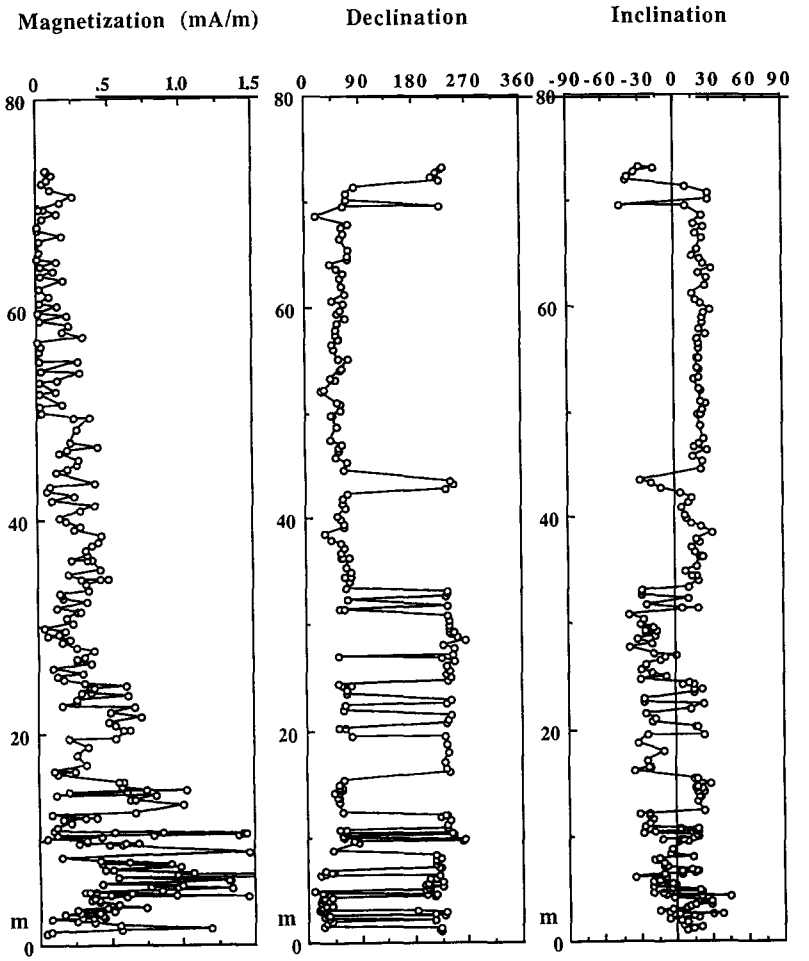


Fig. 9. Magnetic stratigraphy of the Bolücektasi Tepe section showing the variations of intensity of total NRM and changes in the direction of ChRM.

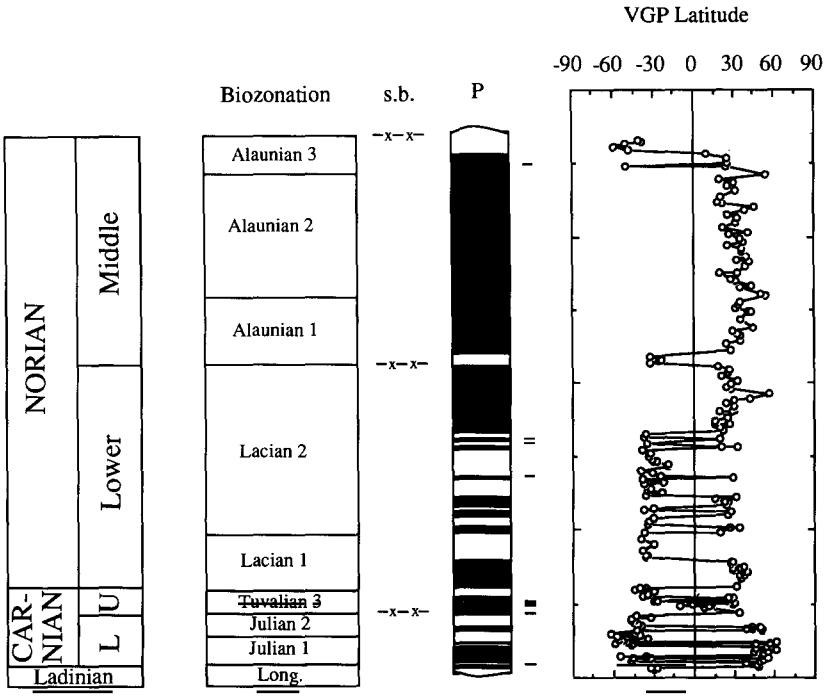


Fig. 10. Magnetic stratigraphy and biozonation of the Bolücektasi Tepe section. The magnetic polarity sequence (column P) is deduced from the VGP latitude column. Magnetic intervals defined by only one sample are indicated by a small horizontal line on the right of column P. We also indicate the major breaks in sedimentation within the section (column s.b.).

We can also use the age estimates for the Triassic stage boundaries derived from the assumption of an equal duration for the ammonoid zones. These estimates, which notably reduce the duration of the Carnian represented in the Bolücektası Tepe section (about 4 million years, but still 7 million years for the Norian), increase the difference in magnetic reversal frequency between the Norian and the Carnian. The frequency becomes 4.3 reversals per million years for the Carnian (2.8 if we exclude the intervals defined by one sample) but remains the same for the Norian (2.9–1.9 reversals per million years). The differences with the previous values illustrate the sensitivity of these estimates to the choice of time scale.

7. Discussion

For 10 years, magnetostratigraphic studies have been carried out for the Middle–Jurassic, Triassic and Late Permian. We have reviewed and selected magnetostratigraphic data which are well correlated to biostratigraphic zonations. Magnetostratigraphic results which are unusable due to an inadequate biostratigraphy are rare (e.g. Channell et al. (1984) for three Early Jurassic Umbrian sections). For the Jurassic, the Hettangian (lowermost Jurassic) is presently undocumented. The Callovian (Middle Jurassic) and the Sinemurian (Early Jurassic) also remain very poorly documented (Channell et al., 1984; Channell et al., 1990). However, the Callovian (with the Oxfordian) is probably represented in the M25–M38 sequence proposed by Handschumacher et al. (1988). If we assume constant seafloor spreading for the Japanese magnetic lineations, the older marine anomalies could be of Upper Bathonian–Lower Callovian age. This sequence of frequent magnetic reversals does not confirm the Jurassic 'quiet zone' suggested in the first M sequences (Larson and Hilde, 1975; Cande et al., 1978; Vogt and Einwich, 1979). Magne-

tostratigraphic results for the Oxfordian are often fragmented and not correlatable with the M sequence (Steiner et al., 1986; Channell et al., 1990). However, these results confirm the high magnetic reversal rate apparent in the Pacific marine magnetic anomaly profiles.

For the Middle Jurassic, the Bajocian is documented by a magnetostratigraphic study of three sections located in the Sub-Betic Cordillera in Spain (Steiner et al., 1987). Despite a high sampling density and a good biostratigraphic control, many polarity intervals are defined by only one sample (about 50% of the intervals in the Carcabuey section) and the different records do not show a clear correlation. In reference to the longer section (Carcabuey section), the reversal frequency can be estimated to range from about 3.5 (excluding the polarity intervals defined by a single sample) to 8 (with all intervals) reversals per million years. The complementary results obtained for the Bathonian (Steiner et al., 1987) are difficult to interpret because the sedimentation rates of the studied sections are lower with respect to the high reversal frequency. Only a minimum estimate of the number of magnetic reversals can be obtained (more than 20 reversals?).

The Lower Bajocian (Middle Jurassic) to Pliensbachian (Lower Jurassic) Breggia section in Switzerland probably provides the best magnetostratigraphic sequence for these periods (Horner and Heller, 1983). More than 60 magnetic reversals are obtained within this section. The polarity intervals, correlated to the ammonite zonations, are often well defined by several samples and this sequence could be a good reference for other studies. However, no reliable correlation is presently obtained with other results. The magnetic sequence from the Italian Alpe Turati section of Domerian age (Late Pliensbachian, Horner and Heller, 1983) shows only a high reversal frequency in agreement with the previous results. For this same period (Pliensbachian), the discrepancy between the Breggia and the Hungarian Bakonycsernye section (Marton et al., 1980) can be explained by a significant difference in the time resolution. The lower magnetic frequency derived from the Hungarian section could be due to a sedimentation rate lower by about one order of magnitude. Two Toarcian sections (Galbrun et al., 1988, 1990), with a well-defined biostratigraphy, can also be compared with the Breggia sequence but this comparison is not satisfactory. Some sedimentary gaps exist within the condensed Toarcian stratotype (Galbrun et al., 1988) and the number of reversals is clearly biassed downward. In contrast, the Toarcian section of Iznalloz in Spain (Galbrun et al., 1990) seems to show a higher number of reversals than the Breggia section, but the intervals are often poorly defined. Moreover, the sampling is not continuous in the lower part of the section due to a marly facies. From the Breggia section and following Harland et al. (1989), we estimate a magnetic frequency of about 2.2 reversals per million years for the Aalenian, 1.7 reversal per million years for the Toarcian and 5.3 reversals per million years for the Pliensbachian. This rapid change in magnetic frequency could be real (the reversal process is highly non-stationary) but it could also be related to a noticeable change in time resolution within the Breggia section. The sedimentation rate changes from about 1 m per million years in the Aalenian to 2 m per million years in the Toarcian and more than 3 m per million years during the Pliensbachian. The higher resolution thus corresponds to the highest magnetic frequency and these results could be interpreted in terms of 'smoothing' of the signal. Only complementary data can provide a definitive answer.

Magnetostratigraphic data for the Triassic have often been obtained on continental sediments in fluvial or lacustrine environments and have insufficient biostratigraphic control (Helsley, 1969; Reeve and Helsley, 1972; Baag and Helsley, 1974a,b; Helsley and Steiner, 1974; Steiner and Helsley, 1974; Larson et al., 1982; Shive et al., 1984; McIntosh et al., 1985; Gallet, 1988; Witte et al., 1991). However, the Early Triassic magnetostratigraphy of marine sediments in China (Steiner et al., 1989) and in the northern Canadian Arctic (Ogg and Steiner, 1991) provides a sequence of numerous reversals. When compared, the two sequences have a similar number of magnetic polarity zones and a magnetostratigraphic correlation can be suggested. In the Chi-

nese sections, many of the magnetic polarity zones are defined by only one sample. Moreover, we note that the directions obtained from these sections are very scattered (see also Heller et al., 1988). Large changes in the magnetic properties occur within these sections and the primary directions are perhaps difficult to isolate. In view of these difficulties, a sequence of 17 magnetic intervals defined by several samples with an 'apparent' good direction can be considered from the 'HPT' section encompassing most of the Early Triassic (column E, Fig. 13; Steiner et al., 1989). This 'filtered' sequence is very similar to the composite Early Triassic magnetostratigraphic sequence suggested by Steiner et al. (1989). However, four of the 18 intervals are poorly defined. By comparison, the Arctic sections (Ogg and Steiner, 1991) provide a composite magnetic polarity sequence of 21 magnetic zones. Again four intervals are poorly defined. Following Harland et al. (1989), with a duration of only 4 million years for the Early Triassic, a magnetic frequency of about 4.5 reversals per million years can be derived (and about 3 reversals per million years considering only the well-defined intervals). However, a longer duration for the Early Triassic is probable (Odin and Odin, 1990) in which case, the magnetic frequency would be largely overestimated. In our estimates using an equal duration for the ammonoid zones, the duration of the

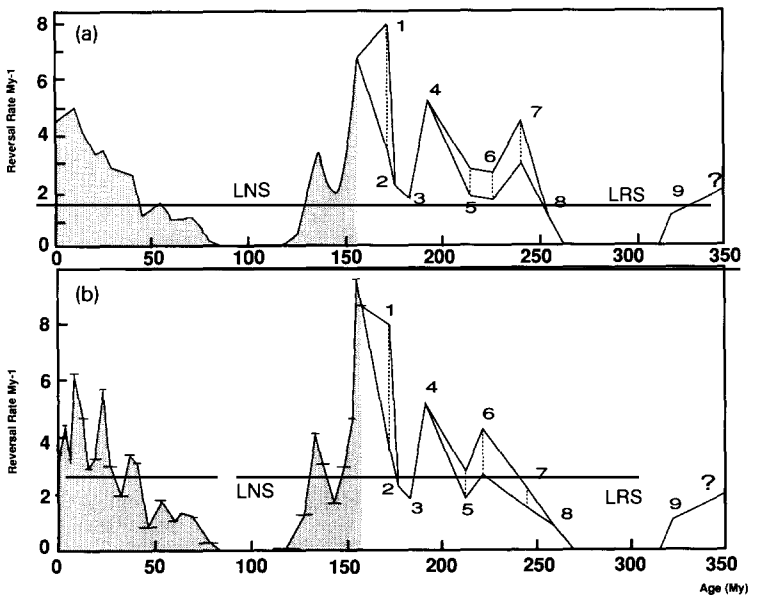


Fig. 11. Changes in the magnetic reversal frequency since 350 Ma. In (a) the shaded area from the Late Jurassic to the present is obtained from the magnetic time scale of Harland et al. (1989), using sliding windows of 10 million years shifted by 5 million years. We also add the M25–M38 sequence of Handschumacher et al. (1988). The magnetic frequencies before 160 Ma are estimated from the magnetostratigraphic data using the Harland et al. time scale. The upper curve connects the frequency estimates found when all magnetic intervals within the sections are considered. The lower curve joins the frequency estimates when only the well-defined magnetic intervals are considered. For each magnetostratigraphic datum, the two frequency estimates are connected by dashed lines. In (b) the magnetic reversal frequencies from the Upper Jurassic to the present are estimated considering time windows defined by the duration of the different geologic stages. We also consider the concept of equal duration of the ammonoid zones to estimate the reversal frequency through the Trias. In these two figures, the magnetostratigraphic data seem to be roughly in agreement with the previously suggested time constant of about 150 million years in the magnetic reversal process. LNS is the long normal superchron, LRS is the Kiaman reverse superchron.

Early Triassic is 8 Ma, giving a reversal frequency of about 2.3 reversals per million years (and 1.5 reversal per million years if the one sample intervals are removed from the sequence). No data are presently available for the Middle Triassic.

Magnetostratigraphic data are also very rare for the Permian even though the Kiaman superchron has been consistently recognized. This interval was recognized by the large predominance of reversed polarity in Permian and Late Carboniferous red beds studied in North America, Australia and Europe (e.g. Graham, 1955; Cox and Doell, 1960; McMahon and Strangway, 1968; Nairn and Peterson, 1973). Only rare polarity sequences, however, clearly contain a continuous fraction of the Kiaman interval (e.g. Gallet (1988) for the Late Permian; Miller and Opdyke (1985) and DiVenere and Opdyke (1990, 1991) for the Late Carboniferous). Recent magnetostratigraphic studies have focussed on the Permo–Triassic boundary (e.g. McFadden et al., 1988; Heller et al., 1988; Steiner et al., 1989; Haag and Heller, 1991). However, the calibration of the end of Kiaman and the number of magnetic intervals before the Permo–Triassic boundary remain uncertain (Steiner et al., 1989; Haag and Heller, 1991). From Chinese and Russian sections, it seems that about three reversals occur within the Upper Tatarian (Steiner et al., 1989). However, reliable magnetostratigraphic results obtained from the Nammal Gorge, Northwest Pakistan (Haag and Heller, 1991), clearly show that the end of Kiaman predates the Upper Tatarian. According to Haag and Heller (1991), a rough frequency of about 1 reversal per million years can be suggested for the Upper Permian.

The magnetic reversal frequency estimates for (1) Bajocian (Steiner et al., 1987), (2) Aalenian, (3) Toarcian, (4) Pliensbachian (Horner and Heller, 1983), (5) Norian (this study), (6) Carnian (this study), (7) Lower Triassic (Steiner et al., 1989; Ogg and Steiner, 1991), (8) Upper Permian (Steiner et al., 1989; Haag and Heller, 1991), and (9) Late Carboniferous (DiVenere and Opdyke, 1990,1991) are plotted together in Figs. 11 (a) and (b) with the frequencies derived from the continuous Cenozoic and Late–Middle Mesozoic magnetic polarity time scale of Harland et al. (1989).

In Fig. 11(a), magnetic reversal frequencies from the continuous geomagnetic sequence were calculated using sliding windows of 10 million years shifted by 5 million years. Moreover, we report the frequencies derived from the M25–M38 sequence of Handschumacher et al. (1988), considering the M25n and M29n intervals as tie-points for a linear scaling. Magnetic frequencies from the magnetostratigraphic data were also estimated from the Harland et al. (1989) time scale. This figure reveals rapid and important changes in magnetic frequency between the Middle Jurassic and the Upper Permian. Despite apparent noise, a rough agreement can be seen with the long-period changes in frequency suggested by McFadden and Merrill (1984), Courtillot and Besse (1987) and Besse and Courtillot (1991). In this model, the magnetic frequency should increase gradually after the Kiaman superchron, from the Upper Permian till the Early–Middle Jurassic, similar to the behavior after the Cretaceous LNS. However, the results for the Early Triassic from China (Steiner et al., 1989), with a high–medium frequency, and for the Aalenian and Toarcian from Breggia (Horner and Heller, 1983), with a low frequency, depart significantly from this behavior. The discrepancy for the Early Triassic could be due to an underestimation of the duration of this period. The rapid changes in frequency during the Early–Middle Jurassic, if significant, are quite similar to the changes observed during the Lower Cretaceous. But, in general, only short time periods can be considered with the available magnetostratigraphic data, and it is not yet possible to perform a complete analysis of superimposed periodicities (Lowrie, 1982; Lowrie and Kent, 1983; Mazaud et al., 1983).

In order to consider roughly a similar smoothing for both continuous and discontinuous data, we have estimated in Fig. 11(b) the magnetic reversal frequencies using successive (somewhat arbitrary) time windows defined by the duration of the different geologic stages. By these new estimates, we show that the general trend in frequency for the Cenozoic and the upper part of the Mesozoic is similar to the previous one. However, short-period fluctuations, close to the suggested 15 million years periodicity (e.g. Mazaud

et al., 1983) are now clearly observed; this pattern is very similar to the one which is obtained by a sliding window of 5 Ma shifted by 2 Ma.

As pointed out previously, different ages and durations can be obtained for the Triassic stages using the concept of equal duration for the ammonoid zones. When compared with the Harland et al. (1989) time scale, the duration of the Late Triassic is reduced by about a factor 1.2 while the duration of the Early Triassic is increased by a factor 2. The new estimates of the magnetic frequency within the Triassic are also plotted in Fig. 11(b) together with unchanged Jurassic magnetostratigraphic data. Again, the trend given by this new curve is reasonably similar to the curve observed in Fig. 11(a). The longer duration of the Early Triassic provides, for this period, a magnetic reversal frequency intermediate between the frequency estimates for the Late Permian and the Late Triassic, in agreement with the suggested long periodicity in the reversal process.

When we compare the fluctuations in frequency between the Cenozoic and the Upper Permian–Late Jurassic period shown in Fig. 11(b), we observe that, although the general pattern is similar, the reversal frequencies are higher in the Early–Middle Mesozoic. We can suggest that this difference is due to a different resolution between the magnetostratigraphic data (with often a bed to bed sampling) and the sequence provided by the magnetic sea-floor anomalies. The problem of the 'missing' short magnetic polarity intervals during the Cenozoic is well known (e.g. Lowrie and Kent, 1983). New short intervals (e.g. 57 intervals as suggested by LaBrecque et al., 1977) would significantly increase the Cenozoic reversal frequencies. Moreover, we note that the reversal frequency range derived from the magnetostratigraphic data decreases significantly when we consider only the well-defined polarity intervals (Fig. 11(b)), being comparable to what is observed during the Cenezoic.

Finally, the comparison between Figs. 11 (a) and (b) clearly shows that, rather than the problem of different smoothings depending on the method of frequency calculation or time resolutions, major limitations for testing the general trend in magnetic reversal frequency since the Kiaman reversed superchron are due to an insufficient absolute chronology of the geologic stage limits. Reliable conclusions will be obtained only by the acquisition of new radiometric and magnetostratigraphic data.

8. Conclusion

The magnetostratigraphy of the Bolücektası Tepe section provides the first Late Triassic magnetic polarity sequence well calibrated with the conodont zonation. A magnetic sequence of 18 polarity intervals for the Carnian and 21 intervals for the Lower–Middle Norian has been defined. A major sedimentary gap including the Middle Carnian and the lower part of the Late Carnian divides the section into two subsections which give two different mean directions. These directions indicate a northward latitudinal movement of $4.8° \pm 2.1°$ and a large change in declination, of about 30°, during this gap. However, no apparent trend is recognized within these two sequences.

We have reviewed the available magnetostratigraphic data for the Middle–Early Jurassic, Triassic and Late Permian to estimate the changes in magnetic reversal frequency since the Kiaman superchron. We conclude that these results agree roughly with a long 150–200 million years time constant in the reversal process, although significant fluctuations in short time intervals cannot be excluded.

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